MTH2005 — Practical 1: Root Finding

Learning objectives

1. Re-familiarise yourself with Python.
2. Use Python to implement and explore root finding with the bisection method, which we have covered in lectures.

Introduction

Welcome to the first practical session of MTH2005, Mathematical Modelling: Theory and Practice. We cover the *theory* part of the module in lectures, these sessions are all about the *practice*. They are vital preparation for the two courseworks and the group project, which together make up 60% of the module's total assessment.

In these sessions we'll be applying methods covered in lectures to solve real problems using Python. The demonstrators and lecturer will be on hand to help if you run into any trouble or have any questions–please just ask!

Recap: Using Python

1. First start Anaconda Navigator - you can find this by typing in the search box.
2. **Wait patiently** for it to finish setting up (observe the progress bar at the bottom of the window) otherwise you may have some unpleasant errors!
3. Scroll down to find 'Jupyter Lab' and click the 'Launch' button. Note, you do not want 'Jupyter Notebook'.
4. When asked what type of file to open, choose 'Terminal'. This will open a Powershell terminal (as we are using Windows machines) where you can interact with the filesystem using commands. We will only see a few basic command-line instructions, but for now...
5. ...we will start an Interactive Python (iPython) session by typing ipython in the Terminal and hitting enter.
6. Note that to exit iPython you press ctrl-d and hit enter.

At the end of the session: In this session we have only used the Interactive Python terminal. To save your session for future reference, type `%save session1`. This will save the commands up to that point in a file called session1.py. You will find it helpful to create a directory in OneDrive called MTH2005 for storing files related to this module.

Very important: For security, you must log off the machine when you finish.

TASK

We will investigate the **bisection method** of root finding. Consider finding the root of the equation

$$f(x) = x^3 - 2x^2 - 3 = 0\,.$$

We start the bisection algorithm with the interval [2,4]. The steps of the bisection method are:

1. Calculate the mid point, c, of the domain $[a, b]$, giving $c = (a + b)/2$.
2. Determine which section of the bisected domain to retain using the Intermediate Value Theorem (IVT).

In the file `practical1.py` (on ELE) is a Python script with six gaps for you to fill in, marked as `GAP1` to `GAP6`. The code will not run properly until these gaps are completed. You should fill in the gaps and check that the code is running as expected.

FURTHER INVESTIGATION

Change the code to fix the number of iterations to 10. Measure the maximum error (see lecture notes) at steps 9 and 10 and check that this is consistent with theory.

The code uses a Python `f-string` to print the results. Remind yourself what this is. How can you format the output so that numbers are printed to a set number of decimal places?